CO1

PROGRAM 1 Date:13/08/26

Aim: Design a login form with username and password using linear layout & Toast valid credentials.

Program:

Mainactivity.java:

```
package com.example.login;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText user,pass;
    Button login;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        user=findViewById(R.id.etusername);
        pass=findViewById(R.id.etpassword);
        login=findViewById(R.id.btlogin);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }

    public void onclick(View view) {
        String s1=user.getText().toString();
        String s2=pass.getText().toString();
        if(s1.equals("basil")&& s2.equals("basil123"))
        {
            Toast.makeText(this, "LOGIN SUCCESSFUL", Toast.LENGTH_SHORT).show();
        }
        else{
            Toast.makeText(this, "INVALID USERNAME OR PASSWORD", Toast.LENGTH_SHORT).show();
        }

    }
}
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"


  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="108dp"
    android:orientation="horizontal">

    <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="User name" />

    <EditText
      android:id="@+id/etusername"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:ems="10"
      android:inputType="text" />
  </LinearLayout>

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="111dp"
    android:orientation="horizontal">

    <TextView
      android:id="@+id/textView2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="Password" />

    <EditText
      android:id="@+id/etpassword"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:ems="10"
      android:inputType="textPassword" />


  </LinearLayout>
  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="112dp"
    android:orientation="horizontal">

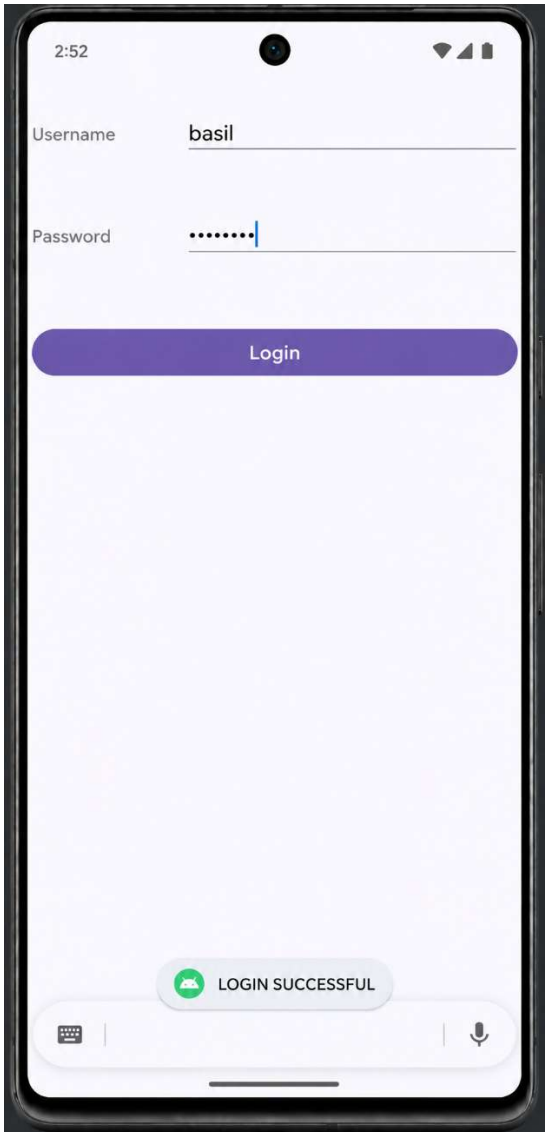

    <Button
      android:id="@+id/btlogin"
      android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:onClick="onclick"



 android:text="Login" />
 </LinearLayout>
</LinearLayout>
```

Output:



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PROGRAM 2 Date:3/09/26

Aim: Write/execute a program that demonstrates activity lifecycle.

Program:

Mainactivity.java:

```
package com.example.lifeCycle;
import...
public class MainActivity extends AppCompatActivity
{
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);

      ViewCompat.setOnApplyWindowInsetsListener
      (findViewById(R.id.main), (v, insets) -> {
         Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
         v.setPadding(systemBars.left, systemBars.top,
               systemBars.right, systemBars.bottom);
         return insets;
      });
   }

   protected void onStart()
   {
      super.onStart();
      Toast.makeText(this, "onStart() stage", Toast.LENGTH_SHORT).show();
   }

   protected void onResume()
   {
      super.onResume();
      Toast.makeText(this, "onResume() stage", Toast.LENGTH_SHORT).show();
   }

   protected void onPause()
   {
      super.onPause();
      Toast.makeText(this, "onPause() stage", Toast.LENGTH_SHORT).show();
   }

   protected void onRestart()
   {
      super.onRestart();
      Toast.makeText(this, "onRestart() stage", Toast.LENGTH_SHORT).show();
   }

   protected void onStop()
   {
      super.onStop();
      Toast.makeText(this, "onStop() stage", Toast.LENGTH_SHORT).show();
   }

   protected void onDestroy()
   {
```

```
        super.onDestroy();
        Toast.makeText(this, "onDestroy() stage", Toast.LENGTH_SHORT).show();


    }
}
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <!-- Username Section -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_vertical"
        android:orientation="horizontal">

        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="Username"
            android:textAppearance="@style/TextAppearance.AppCompat.Body1" />

        <EditText
            android:id="@+id/userid"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="2"
            android:ems="10"
            android:inputType="text"
            android:minHeight="48dp"
            android:textAppearance="@style/TextAppearance.AppCompat.Body1" />
    </LinearLayout>

    <!-- Password Section -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:gravity="center_vertical"
        android:orientation="horizontal">
```

```
        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"


    android:text="Password"
    android:textAppearance="@style/TextAppearance.AppCompat.Body1" />

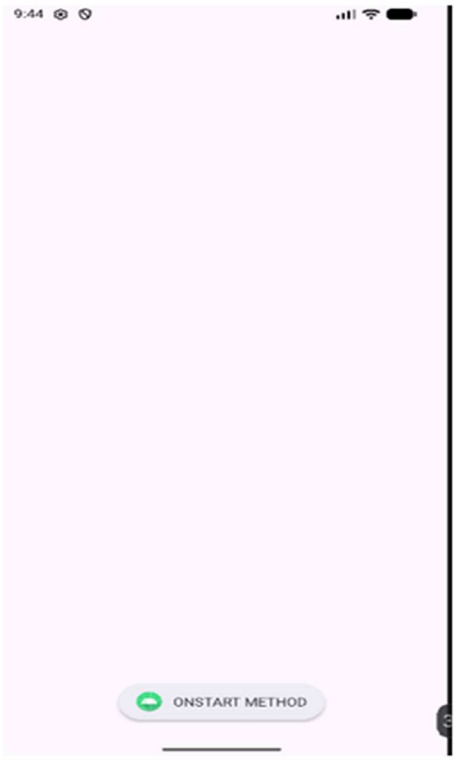
        <EditText
            android:id="@+id/passid"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="2"
            android:ems="10"
            android:inputType="textPassword"
            android:minHeight="48dp" />
    </LinearLayout>

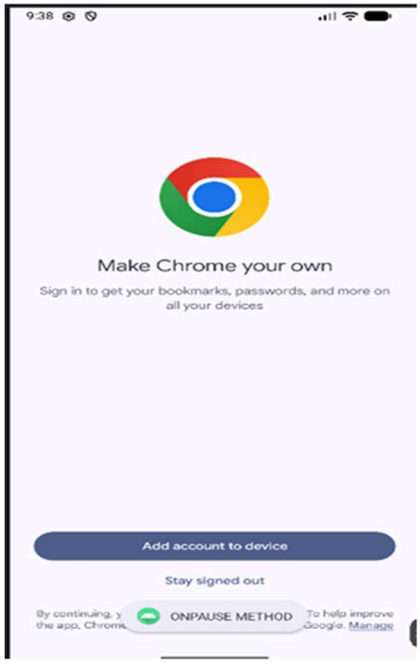
    <!-- Login Button -->
    <Button
        android:id="@+id/btid"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:onClick="onClick"
        android:text="Login"
        android:textAppearance="@style/TextAppearance.AppCompat.Body1" />

</LinearLayout>
```

Output:


9:37
ONRESUME METHOD


9:44
ONSTART METHOD


9:38
Make Chrome your own
Sign in to get your bookmarks, passwords, and more on all your devices
Add account to device
Stay signed out
ONPAUSE METHOD

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PROGRAM 3 Date:1/10/26

Aim: Implement validations on various UI controls .

Program:

Mainactivity.java:

```
package com.example.validation;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText e_name, e_email, e_pwd;
    RadioGroup rg_gender;
    RadioButton r_female;
    CheckBox c_agree;

    boolean isAllFieldsChecked = false;

    @SuppressLint({"WrongViewCast", "MissingInflatedId"})
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Connect XML views with Java
        e_name = findViewById(R.id.e_name);
        e_email = findViewById(R.id.e_email);
        e_pwd = findViewById(R.id.e_pwd);

        rg_gender = findViewById(R.id.rg_gender);
        r_female = findViewById(R.id.r_female);
        //r_male = findViewById(R.id.r_male);
        //r_other = findViewById(R.id.r_other);

        c_agree = findViewById(R.id.c_agree);
    }

    public void submit(View view) {

        isAllFieldsChecked = CheckAllFields();

        if (isAllFieldsChecked) {
            Toast.makeText(this, "Successfully", Toast.LENGTH_SHORT).show();
        }
    }

    private boolean CheckAllFields() {

        // Check Name
```

```
if (e_name.length() == 0) {
            e_name.setError("This field is required");
            e_name.requestFocus();
            return false;
        }

        // Check Email
        if (e_email.length() == 0) {
            e_email.setError("Email is required");
            e_email.requestFocus();
            return false;
        } else {

            String s_email = e_email.getText().toString();

            String checkEmail = "[a-zA-Z0-9._-]+@[a-z]+.+[a-z]+";

            if (!s_email.matches(checkEmail)) {
                e_email.setError("Invalid Email");
                e_email.requestFocus();
                return false;
            }
        }

        // Check Password
        if (e_pwd.length() == 0) {
            e_pwd.setError("Password is required");
            e_pwd.requestFocus();
            return false;

        } else if (e_pwd.length() < 8) {
            e_pwd.setError("Password must be minimum 8 characters");
            e_pwd.requestFocus();
            return false;
        }

        // Check Gender
        if (rg_gender.getCheckedRadioButtonId() == -1) {
            r_female.setError("Select Gender");
            r_female.requestFocus();
            return false;
        }

        // Check Terms and Conditions
        if (!c_agree.isChecked()) {
            c_agree.setError("Please agree before submitting");
            c_agree.requestFocus();
            return false;
        }

        // All fields are valid
        return true;
    }
}
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/scrollView"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:background="#F4F8FF"
 android:fillViewport="true"
 tools:context=".MainActivity">

 <LinearLayout
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">

    <!-- Blue B Logo -->
    <TextView
       android:layout_width="match_parent"
       android:layout_height="120dp"
       android:gravity="center"
       android:text="B"
       android:textColor="#087BEE"
       android:textSize="76sp"
       android:textStyle="bold"
       android:fontFamily="sans-serif-black"
       android:contentDescription="B logo" />

    <!-- Blue Registration Header -->
    <TextView
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:background="#147FF5"
       android:gravity="center"
       android:padding="16dp"
```

```
android:text="Enter the details for registration"
    android:textColor="#FFFFFF"
    android:textSize="18sp" />

<!-- Registration Form -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp">

    <!-- Name -->
    <EditText
        android:id="@+id/e_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:hint="Name"
        android:inputType="textPersonName"
        android:minHeight="48dp"
        android:text="basil"
        android:textColor="#101D40"
        android:textSize="18sp"
        android:backgroundTint="#1680FF" />

    <!-- Email -->
    <EditText
        android:id="@+id/e_email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:hint="Email"
```

```
 android:inputType="textEmailAddress"
    android:minHeight="48dp"
    android:text="basil123@gmail.com"
    android:textColor="#101D40"
    android:textSize="18sp"
    android:backgroundTint="#1680FF" />

<!-- Gender Label -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="24dp"
    android:text="Gender"
    android:textColor="#101D40"
    android:textSize="16sp"
    android:textStyle="bold" />

<!-- Gender Options -->
<RadioGroup
    android:id="@+id/rg_gender"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical">

    <RadioButton
        android:id="@+id/r_male"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:buttonTint="#147FF5"
        android:checked="true"
```

```
            android:text="Male"
            android:textColor="#101D40" />

        <RadioButton
            android:id="@+id/r_female"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1.2"
            android:buttonTint="#147FF5"
            android:text="Female"
            android:textColor="#101D40" />

        <RadioButton
            android:id="@+id/r_other"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:buttonTint="#147FF5"
            android:text="Other"
            android:textColor="#101D40" />

    </RadioGroup>

    <!-- Password -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:gravity="center_vertical"
        android:orientation="horizontal">

        <TextView
            android:layout_width="100dp"
```

```
        android:layout_height="wrap_content"
        android:text="Password"
        android:textColor="#53668A"
        android:textSize="16sp" />

    <EditText
        android:id="@+id/e_pwd"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:textColor="#101D40"
        android:textSize="18sp"
        android:backgroundTint="#1680FF" />

</LinearLayout>

<!-- Terms and Conditions -->
<CheckBox
    android:id="@+id/c_agree"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="28dp"
    android:buttonTint="#147FF5"
    android:checked="true"
    android:text="Agree Terms and Conditions"
    android:textColor="#101D40"
    android:textSize="14sp" />

<!-- Blue Rounded Register Button -->
<Button
```

```
    android:id="@+id/btn_register"
    android:layout_width="match_parent"
    android:layout_height="56dp"
    android:layout_marginTop="36dp"
    android:backgroundTint="#147FF5
    android:background="@drawable/btn_register_shape"
    android:onClick="submit"
    android:text="Register"
    android:textAllCaps="false"
    android:textColor="#FFFFFF"
    android:textSize="18sp" />

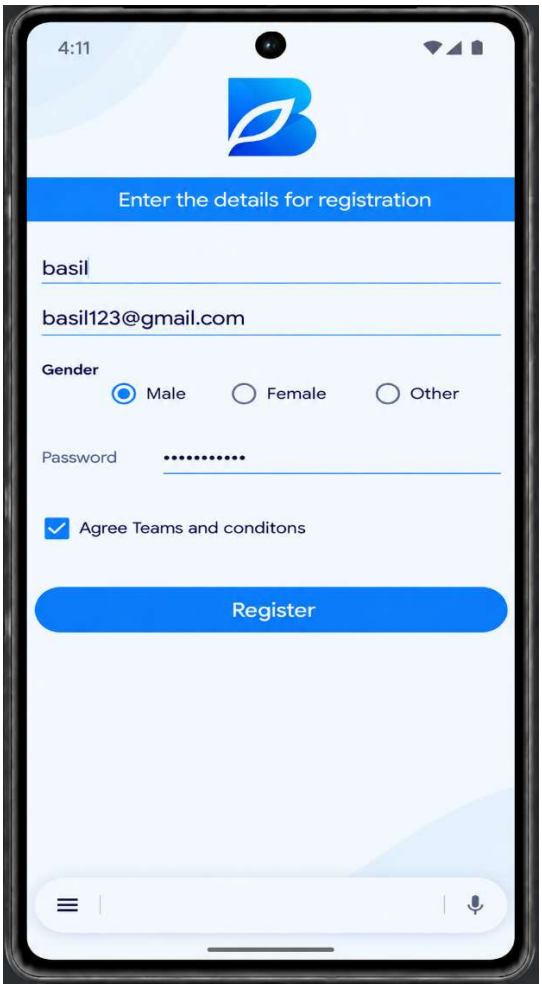
<!-- Student Footer -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="120dp"
    android:layout_marginBottom="32dp"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="🎓"
        android:textSize="30sp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:text="Made by a student"
```

```
            android:textColor="#315B9B"
            android:textSize="16sp" />
        </LinearLayout>
    </LinearLayout>
</LinearLayout>
</ScrollView>
```

Output:



CO2

PROGRAM 1 Date:24/09/26

Aim: Program for calculator using grid layout.

Program:

Mainactivity.java:

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity
        implements View.OnClickListener {

    TextView number;

    double num1 = 0;
    double num2 = 0;

    String op = "";

    Button b1, b2, b3, b4, b5, b6, b7, b8, b9, b0;
    Button sum, sub, mul, div, deci, equal, clear;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        number = findViewById(R.id.et_number);

        // Number buttons
        b1 = findViewById(R.id.b1);
        b2 = findViewById(R.id.b2);
        b3 = findViewById(R.id.b3);
        b4 = findViewById(R.id.b4);
        b5 = findViewById(R.id.b5);
        b6 = findViewById(R.id.b6);
        b7 = findViewById(R.id.b7);
        b8 = findViewById(R.id.b8);
        b9 = findViewById(R.id.b9);
        b0 = findViewById(R.id.b0);

        // Operator buttons
        sum = findViewById(R.id.add);
        sub = findViewById(R.id.subtract);
        mul = findViewById(R.id.multiply);
        div = findViewById(R.id.divide);
        deci = findViewById(R.id.decimal);

        equal = findViewById(R.id.equal);
        clear = findViewById(R.id.delete);
```

```
        // Set click listeners
        b1.setOnClickListener(this);
        b2.setOnClickListener(this);



        b3.setOnClickListener(this);
        b4.setOnClickListener(this);
        b5.setOnClickListener(this);
        b6.setOnClickListener(this);
        b7.setOnClickListener(this);
        b8.setOnClickListener(this);
        b9.setOnClickListener(this);
        b0.setOnClickListener(this);

        sum.setOnClickListener(this);
        sub.setOnClickListener(this);
        mul.setOnClickListener(this);
        div.setOnClickListener(this);
        deci.setOnClickListener(this);

        equal.setOnClickListener(this);
        clear.setOnClickListener(this);
    }

    @SuppressLint("SetTextI18n")
    @Override
    public void onClick(View view) {

        Button b = (Button) view;

        // Clear button
        if (view == clear) {

            number.setText("");
            num1 = 0;
            num2 = 0;
            op = "";

        }

        // Decimal button
        else if (view == deci) {

            String s = number.getText().toString();

            if (!s.contains(".")) {
                number.setText(s + ".");
            }

        }

        // Operator buttons
        else if (view == sum || view == sub ||
                view == mul || view == div) {

            if (!number.getText().toString().isEmpty()) {

                num1 = Double.parseDouble(
                        number.getText().toString()
                );

                op = b.getText().toString();
```

```
            number.setText("");
        }
    }

    // Equal button
    else if (view == equal) {


 if (!number.getText().toString().isEmpty()
            && !op.isEmpty()) {

            num2 = Double.parseDouble(
                number.getText().toString()
            );

            double result = 0;

            if (op.equals("+")) {
                result = num1 + num2;
            }

            else if (op.equals("-")) {
                result = num1 - num2;
            }

            else if (op.equals("*") || op.equals("×")) {
                result = num1 * num2;
            }

            else if (op.equals("/") || op.equals("÷")) {

                if (num2 == 0) {
                    number.setText("Cannot divide by zero");
                    return;
                }

                result = num1 / num2;
            }

            number.setText(String.valueOf(result));

            op = "";
        }
    }

    // Number buttons
    else {

        String s = number.getText().toString();

        number.setText(s + b.getText().toString());
    }
  }
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
    android:padding="12dp"
    tools:context=".MainActivity">

    <!-- Calculator Display Screen -->
    <TextView
        android:id="@+id/et_number"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:layout_marginBottom="24dp"

        android:gravity="end"
        android:padding="16dp"
        android:textColor="#000000"


        android:textSize="32sp" />

    <!-- Row 1: 7, 8, 9, / -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <Button
            android:id="@+id/b7"
            android:layout_width="0dp"
            android:layout_height="64dp"
            android:layout_margin="4dp"
            android:layout_weight="1"
            android:text="7"
            android:textSize="22sp" />

        <Button
            android:id="@+id/b8"
            android:layout_width="0dp"
            android:layout_height="64dp"
            android:layout_margin="4dp"
            android:layout_weight="1"
            android:text="8"
            android:textSize="22sp" />

        <Button
            android:id="@+id/b9"
            android:layout_width="0dp"
            android:layout_height="64dp"
            android:layout_margin="4dp"
            android:layout_weight="1"
            android:text="9"
            android:textSize="22sp" />

        <Button
            android:id="@+id/divide"
            android:layout_width="0dp"
            android:layout_height="64dp"
            android:layout_margin="4dp"
            android:layout_weight="1"
            android:text="/"
            android:textSize="22sp" />
    </LinearLayout>

    <!-- Row 2: 4, 5, 6, * -->
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/b4"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="4"
        android:textSize="22sp" />


<Button
        android:id="@+id/b5"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"



        android:layout_weight="1"
        android:text="5"
        android:textSize="22sp" />

    <Button
        android:id="@+id/b6"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="6"
        android:textSize="22sp" />

    <Button
        android:id="@+id/multiply"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="*"
        android:textSize="22sp" />
</LinearLayout>

<!-- Row 3: 1, 2, 3, + -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/b1"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="1"
        android:textSize="22sp" />

    <Button
```

```
        android:id="@+id/b2"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="2"
        android:textSize="22sp" />

    <Button
        android:id="@+id/b3"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="3"
        android:textSize="22sp" />




<Button
        android:id="@+id/add"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="+"
        android:textSize="22sp" />
  </LinearLayout>



   <!-- Row 4: 0, ., =, - -->
   <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/b0"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="0"
        android:textSize="22sp" />

    <Button
        android:id="@+id/decimal"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
        android:text="."
        android:textSize="22sp" />

    <Button
        android:id="@+id/equal"
        android:layout_width="0dp"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:layout_weight="1"
```

```
            android:text="="
            android:textSize="22sp" />

        <Button
            android:id="@+id/subtract"
            android:layout_width="0dp"
            android:layout_height="64dp"
            android:layout_margin="4dp"
            android:layout_weight="1"
            android:text="-"
            android:textSize="22sp" />
    </LinearLayout>

    <!-- Clear / Delete Button -->
    <Button
        android:id="@+id/delete"
        android:layout_width="match_parent"
        android:layout_height="64dp"
        android:layout_margin="4dp"
        android:text="Del"
        android:textAllCaps="false"
        android:textSize="22sp" />

</LinearLayout>
```

Output:



CO2

PROGRAM 2 Date:28/09/26

AIM: Create a facebook Page using RelativeLayout and XML file.

Program:

Mainactivity.java:

```
package com.example.fb_login_pg;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;


import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   EditText user,pass;
   Button login;


   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      user = findViewById(R.id.editTextTextEmailAddress);
            pass=findViewById(R.id.editTextTextPassword);
            login=findViewById(R.id.button);

      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
         Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
         v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
         return insets;
      });
   }
   public void onClick(View view) {
      String s1=user.getText().toString();
      String s2=pass.getText().toString();
      if(s1.equals("basil")&& s2.equals("basil123"))
```

```
        {
            Toast.makeText(this, "LOGIN SUCCESSFUL", Toast.LENGTH_SHORT).show();
        }
        else{
            Toast.makeText(this, "INVALID USERNAME OR PASSWORD", Toast.LENGTH_SHORT).show();
        }
    }
}
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center_horizontal"
    android:orientation="vertical"
    android:padding="24dp"
    tools:context=".MainActivity">

    <!-- Facebook Icon / Logo placeholder -->
    <!-- Facebook Icon / Logo placeholder -->
    <ImageView
        android:id="@+id/imageView"
        android:layout_width="72dp"
        android:layout_height="72dp"
        android:layout_marginTop="32dp"
        android:contentDescription="Facebook Logo"
        android:src="@android:drawable/ic_menu_share" />


    <!-- Title Header -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="start"
        android:layout_marginTop="24dp"
        android:layout_marginBottom="16dp"
        android:text="Log in to Facebook"
        android:textColor="#000000"
        android:textSize="22sp"
        android:textStyle="bold" />
```

```
<!-- Email / Username Input Field -->


<EditText
   android:id="@+id/editTextTextEmailAddress"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:layout_marginTop="8dp"
   android:hint="Mobile number or email address"
   android:inputType="textEmailAddress"
   android:minHeight="48dp"
   android:textSize="18sp" />

<!-- Password Input Field -->

<EditText
   android:id="@+id/editTextTextPassword"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:layout_marginTop="12dp"
   android:hint="Password"
   android:inputType="textPassword"
   android:minHeight="48dp"
   android:textSize="18sp" />


<!-- Log In Button -->

<Button

   android:id="@+id/button"
   android:layout_width="match_parent"
   android:layout_height="56dp"

   android:layout_marginTop="24dp"

   android:backgroundTint="#1877F2"

     android:onClick="onClick" android:text="Log

     in" android:textAllCaps="false"

    android:textSize="16sp" />



<!-- Forgotten Password Button -->

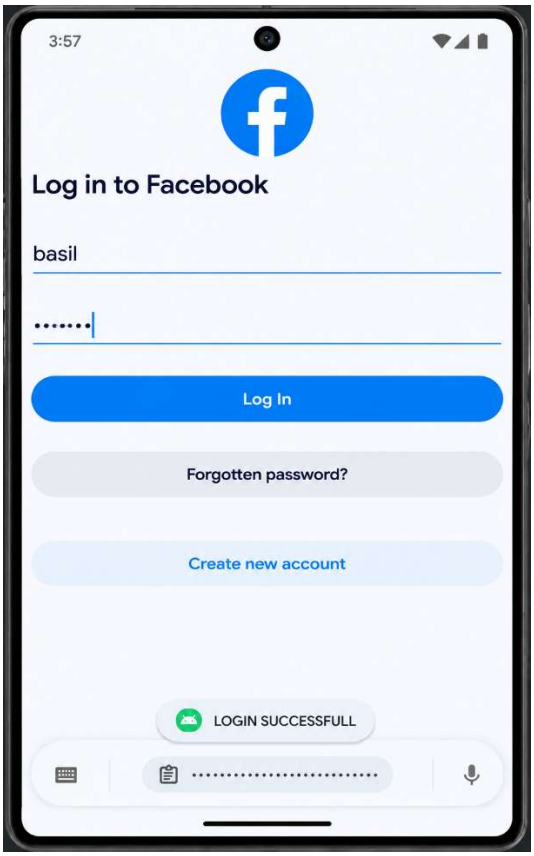
<Button

   android:id="@+id/button_forgot"
   style="?attr/materialButtonOutlinedStyle"
   android:layout_width="match_parent"
   android:layout_height="56dp"
   android:layout_marginTop="16dp"
   android:backgroundTint="#E4E6EB"
   android:text="Forgotten password?"
   android:textAllCaps="false"
   android:textColor="#050505"
   android:textSize="15sp" />
```

```
<!-- Create New Account Button -->
<Button
    android:id="@+id/button_create"
    style="?attr/materialButtonOutlinedStyle"
    android:layout_width="match_parent"
    android:layout_height="56dp"
    android:layout_marginTop="16dp"
    android:backgroundTint="#E4E6EB"
    android:text="Create new account"
    android:textAllCaps="false"
    android:textColor="#1877F2"
    android:textSize="15sp" />

</LinearLayout>
```

Output:



CO2

PROGRAM 3 Date:17/09/26

Aim: Program to toggle an Image.

Program:

Mainactivity.java:

```
package com.example.alanapp;

import android.os.Bundle;
import android.view.View;
import android.widget.ImageView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    ImageView img1,img2;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        img1=findViewById(R.id.image1);
        img2=findViewById(R.id.image2);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }

    public void onClick(View view) {
        if(view==img1){
            img1.setVisibility(View.GONE);
            img2.setVisibility(View.VISIBLE);
        } else {
            img2.setVisibility(View.GONE);
            img1.setVisibility(View.VISIBLE);
        }
    }
}
```

Activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <!-- First Image (Initially Visible) -->
    <ImageView
        android:id="@+id/image1"
        android:layout_width="match_parent"
        android:layout_height="350dp"
        android:clickable="true"
        android:focusable="true"
        android:onClick="onClick"
        android:scaleType="centerCrop"
        android:src="@drawable/image1"
        android:visibility="visible" />

    <!-- Second Image (Initially Hidden) -->
    <ImageView
        android:id="@+id/image2"
        android:layout_width="match_parent"
        android:layout_height="350dp"
        android:clickable="true"
        android:focusable="true"
        android:onClick="onClick"
        android:scaleType="centerCrop"
        android:src="@drawable/image2"
        android:visibility="gone" />

</LinearLayout>
```

Output:

